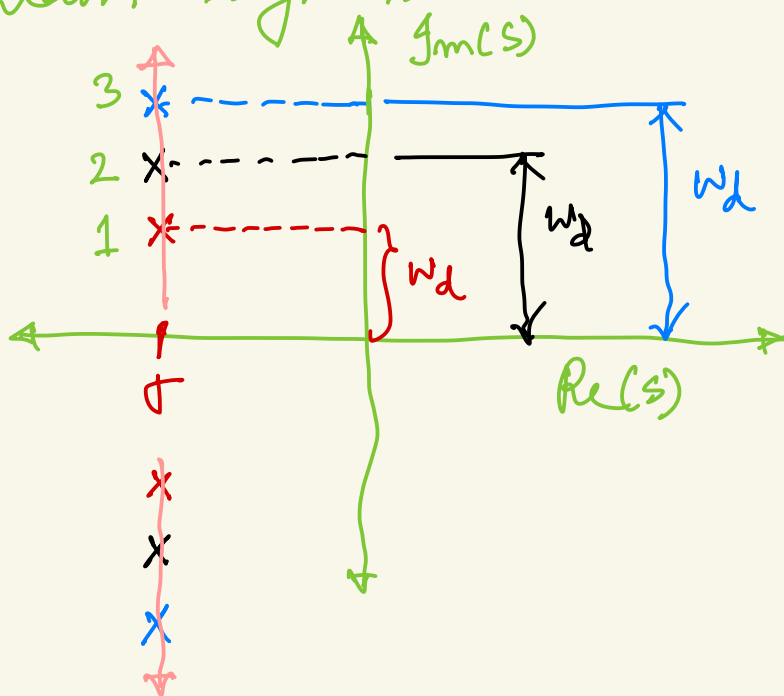




Lecture 22

In this lecture, we would like to briefly investigate the effect of pole locations on the time domain signals.



In the above figure, we keep the real part (σ) constant, but let

the imaginary part ω_d increase

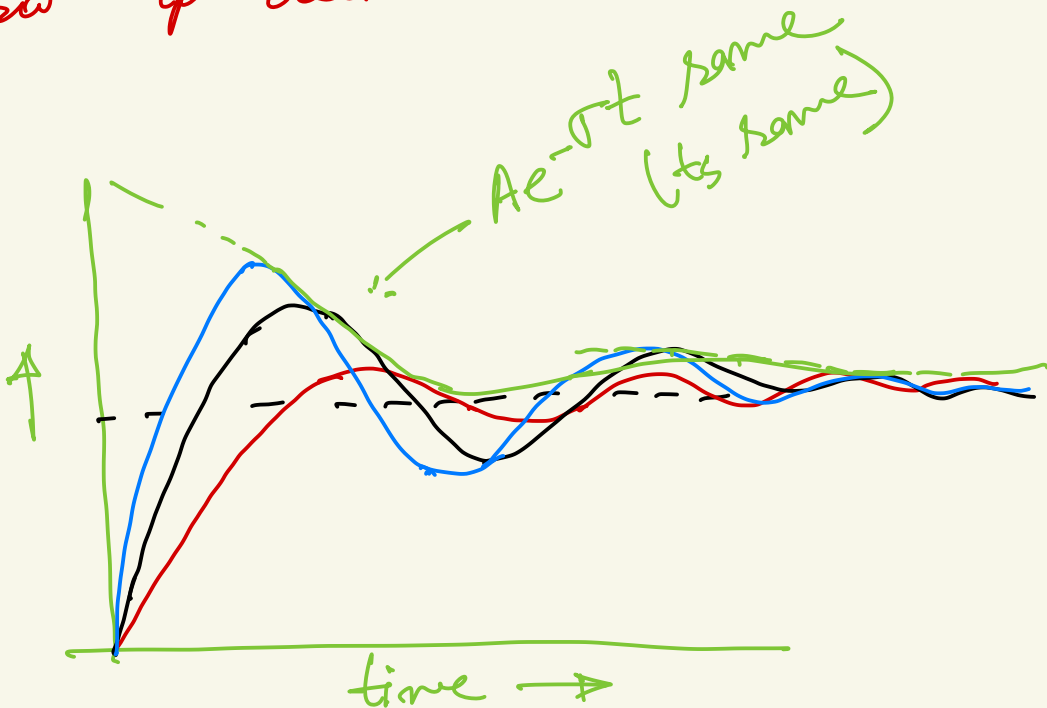
We note that

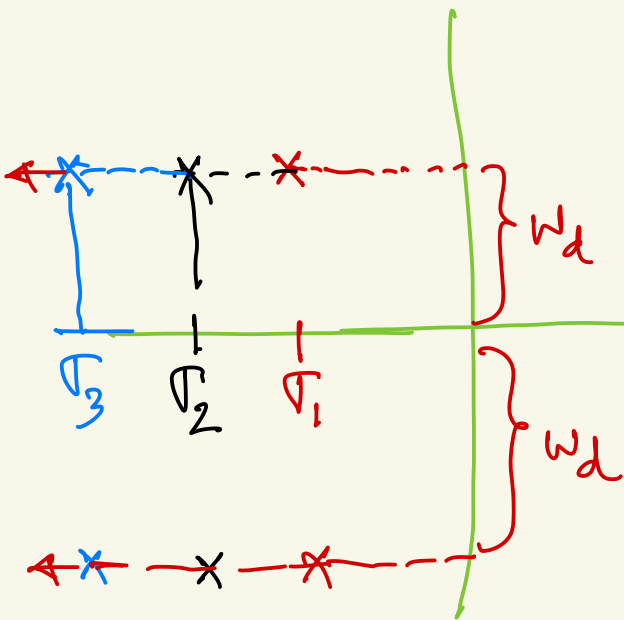
$$3 \quad t_s = \frac{4.3}{\sigma} (\text{constant})$$

so t_s remains constant

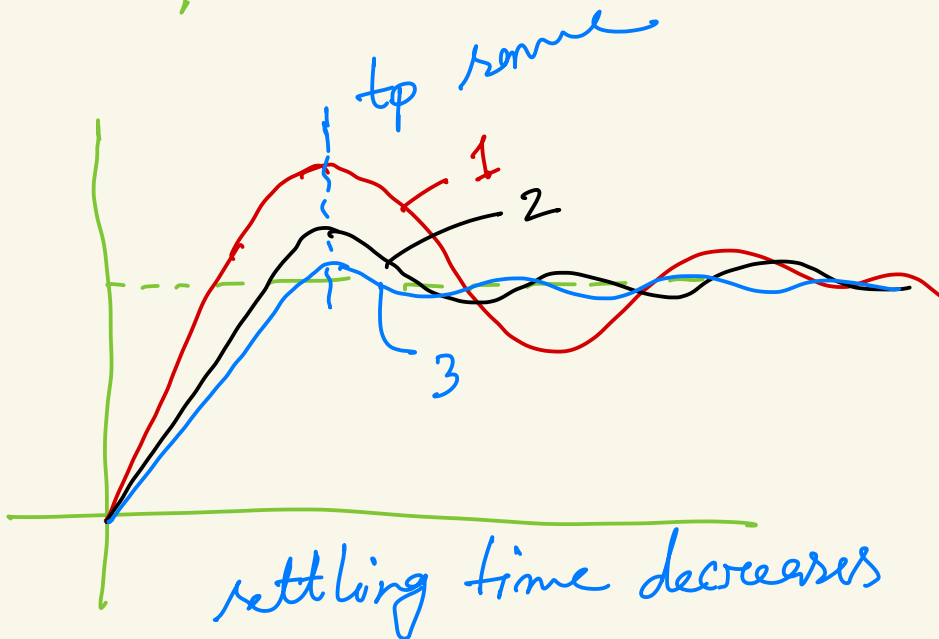
$$\downarrow t_p = \frac{\pi}{\omega_d} \uparrow (\text{increase})$$

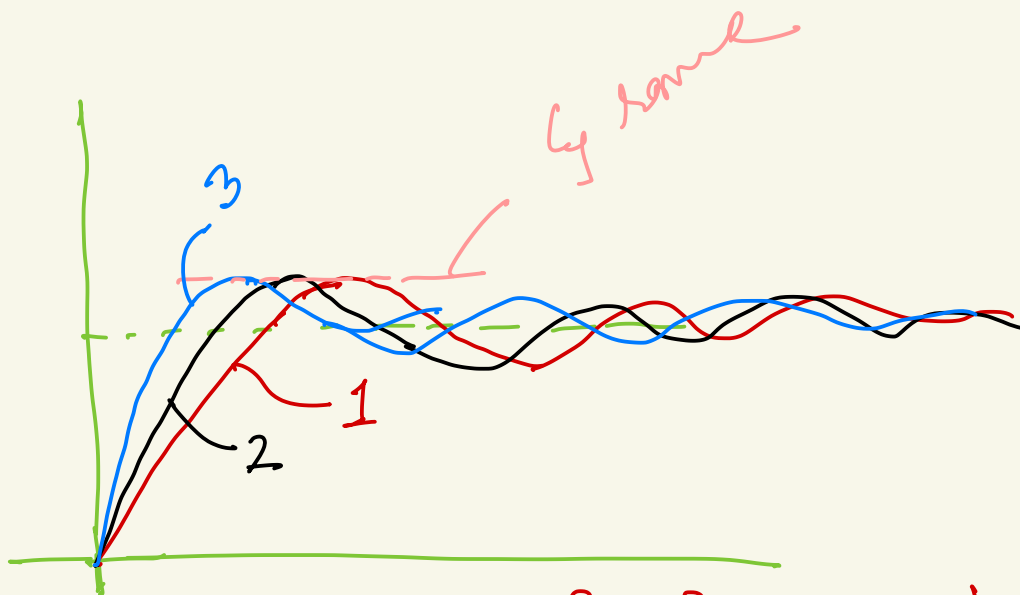
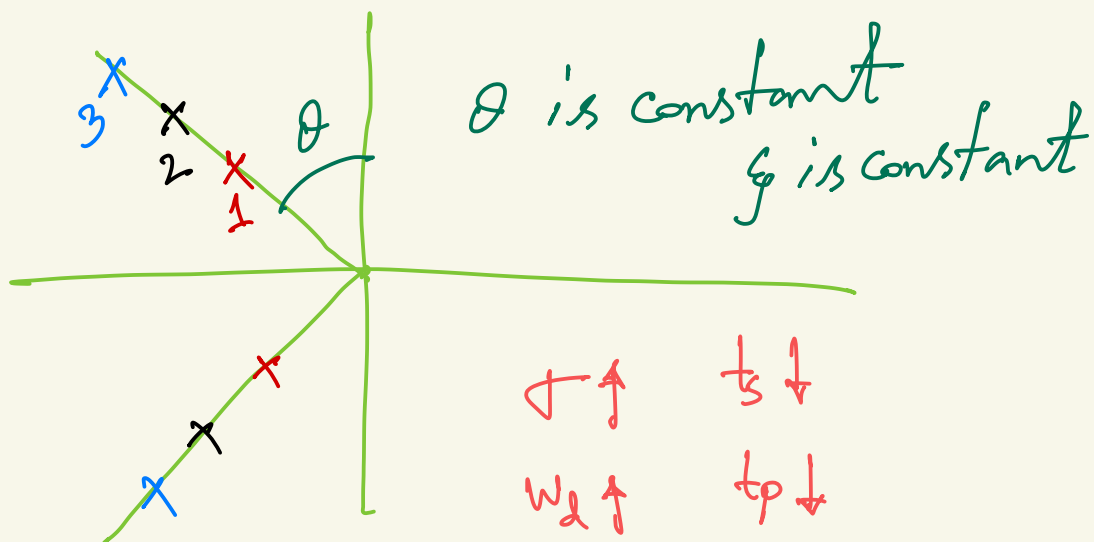
so t_p decreases.





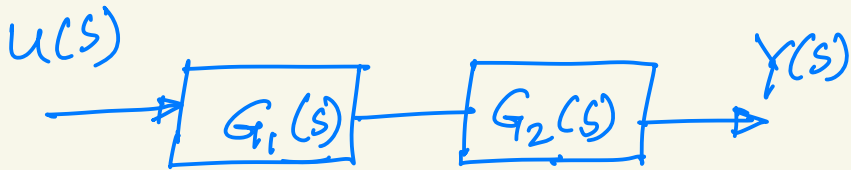
Here ζ increases
but ω_d remains
constant





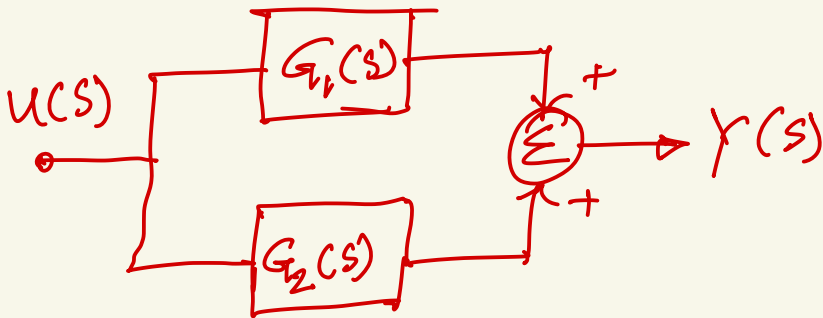
time response of the associated pole.

Block diagrams :



$$\frac{Y(s)}{U(s)} = G_1(s) G_2(s)$$

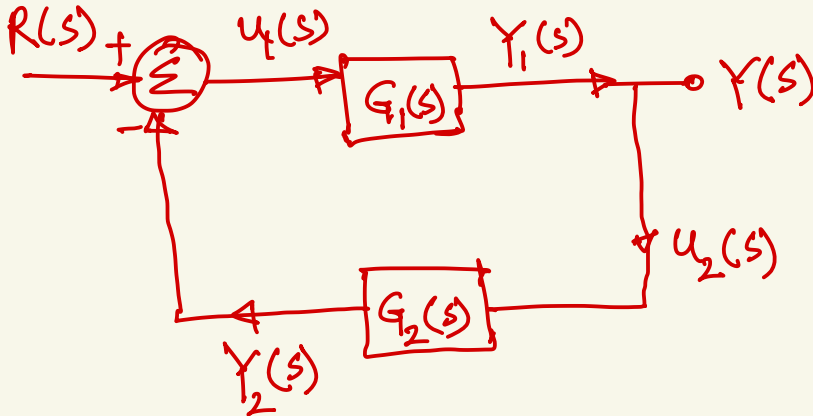
Cascade form



$$Y(s) = U(s)G_1(s) + U(s)G_2(s)$$

$$Y(s) = U(s)(G_1(s) + G_2(s))$$

$$\frac{Y(s)}{U(s)} = G_1(s) + G_2(s)$$



$$U_1(s) = R(s) - Y_2(s)$$

$$Y_1(s) = G_1(s) U_1(s)$$

$$= G_1(s) (R(s) - Y_2(s)) \quad U_2(s) = Y_1(s)$$

$$= G_1(s) (R(s) - G_2(s) U_2(s))$$

$$= G_1(s) (R(s) - G_2(s) Y_1(s))$$

$$Y_1(s) = G_1(s) R(s) - G_1(s) G_2(s) Y_1(s)$$

$$\Rightarrow Y_1(s) [1 + G_1(s)G_2(s)] = G_1(s)R(s)$$

$$\Rightarrow \frac{Y_1(s)}{R(s)} = \frac{G_1(s)}{1 + G_1(s)G_2(s)}$$

This is the closed loop transfer function of the feedback system.